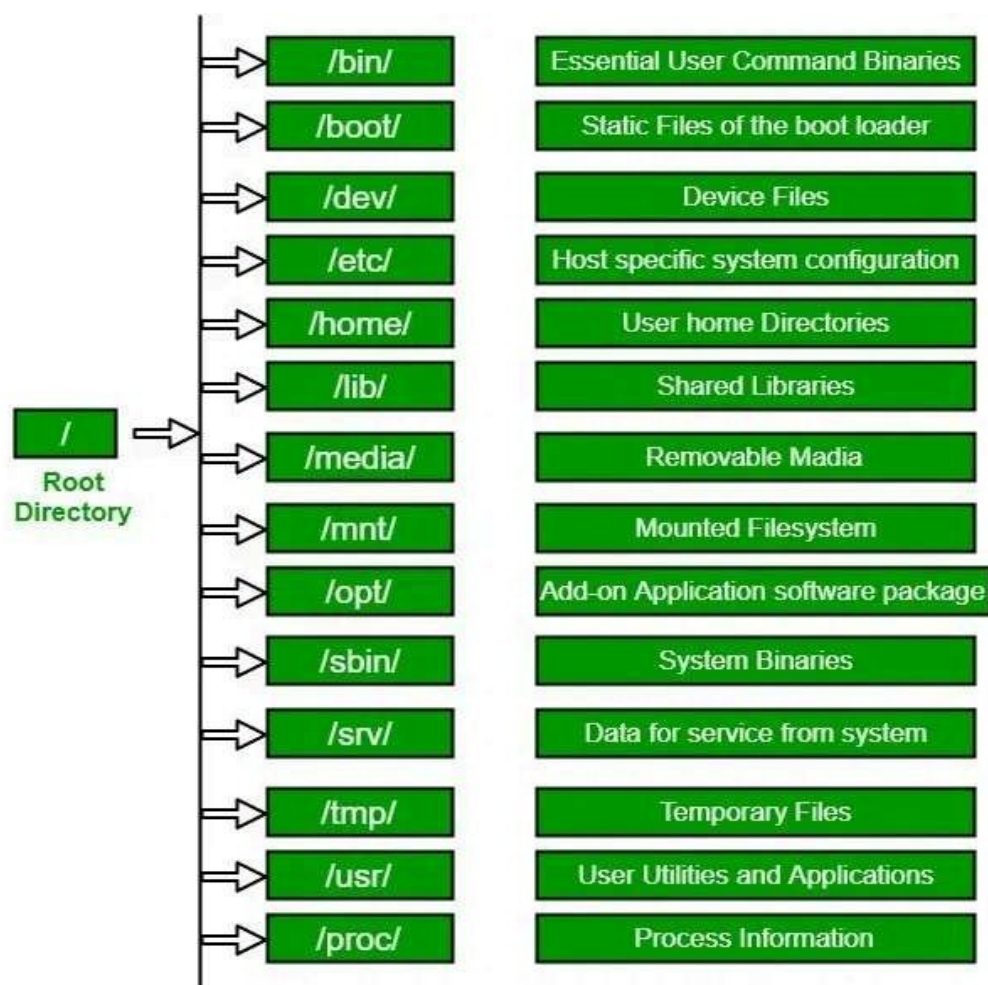


Linux File Hierarchy Structure

The **Linux File Hierarchy Structure** or the **Filesystem Hierarchy Standard (FHS)** defines the directory structure and directory contents in Unix-like operating systems.

It is maintained by the Linux Foundation.

- In the FHS, all files and directories appear under **the root directory /**, even if they are stored on different physical or virtual devices.
- Some of these directories only exist on a particular system if certain subsystems, such as the X Window System, are installed.
- Most of these directories exist in all UNIX operating systems and are generally used in much the same way; however, the descriptions here are those used specifically for the FHS and are not considered authoritative for platforms other than Linux.



1. / (Root):

At the top of every Linux file system is the **root directory** represented by a forward slash **/**.

It's the base point, and [no directory exists above it](#).

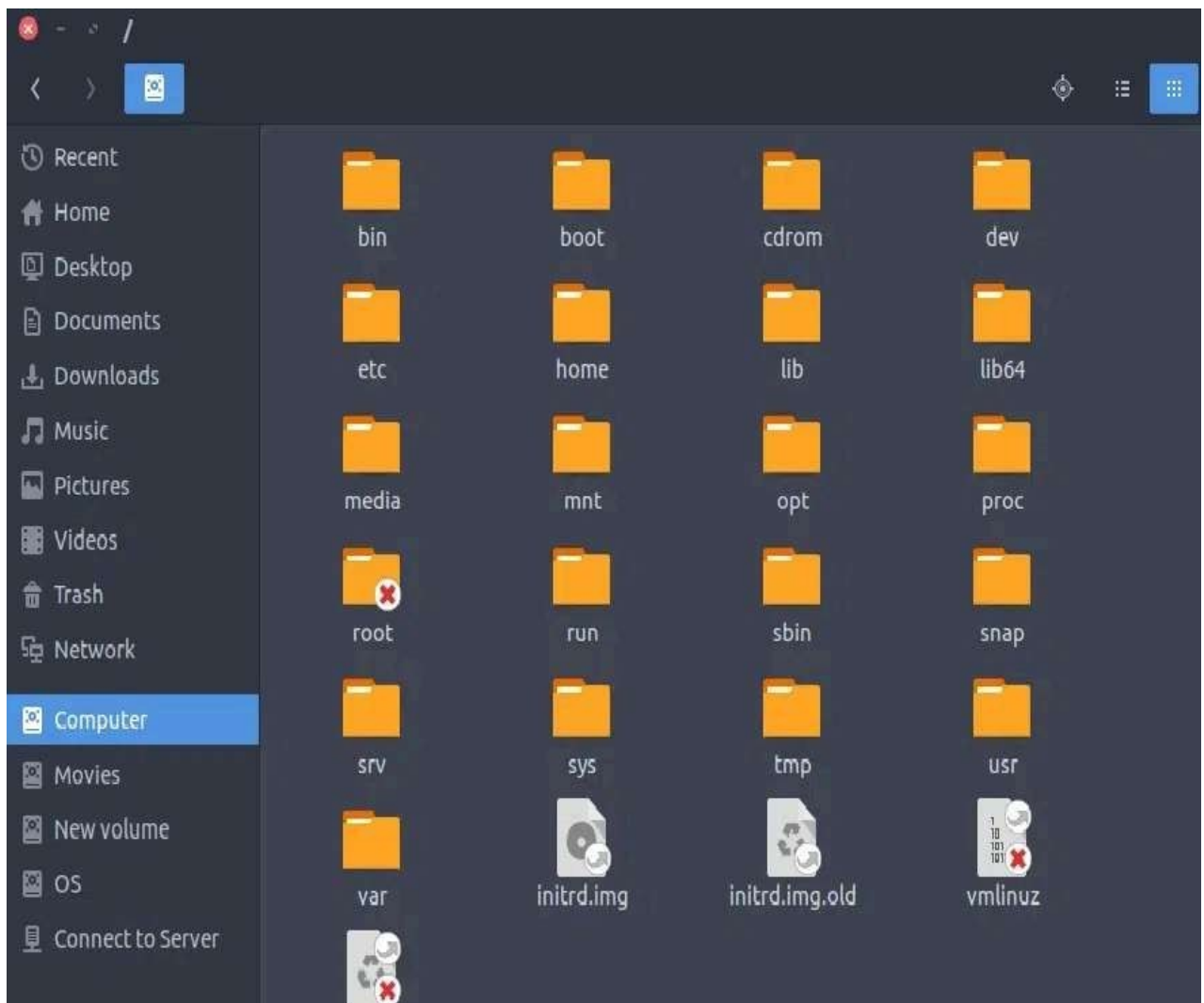
[All other directories branch](#) out from this root directory.

- Every single file and directory start from the root directory.
- [Only root user has the right to write](#) under this directory.
- `/root` is the root user's home directory, which is [not the same as /](#)

[Only the root user has permission to modify contents](#) inside this directory.

Regular users cannot make changes here.

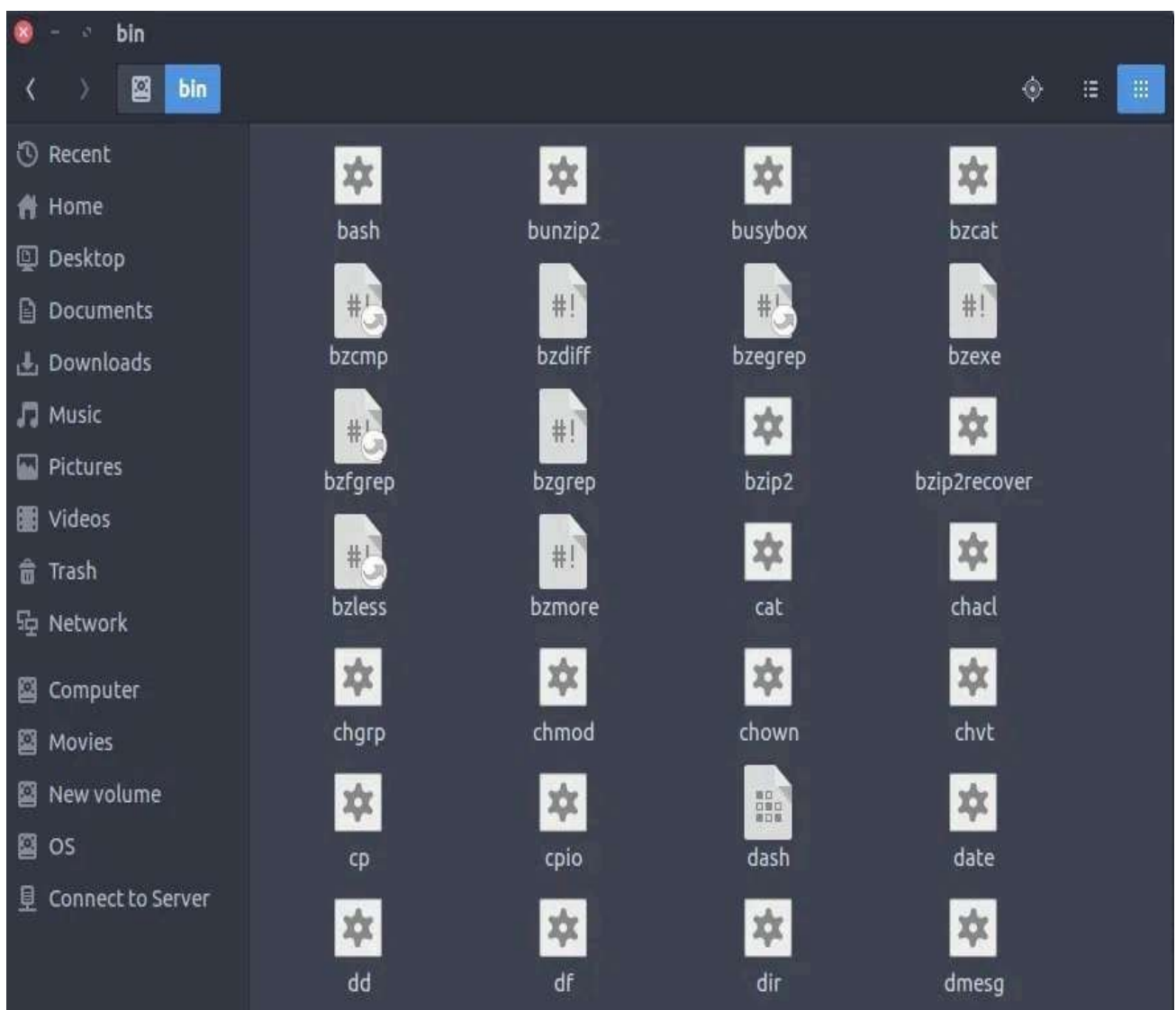
For example, if you attempt to create a file in `/` as a non-root user, you'll encounter a "Permission Denied" error.



2. /bin :

The /bin directory contains essential commands and binaries needed by all users, including `cp`, `ls`, `ssh`, and `kill`. These commands are universally available across user types.

- Contains binary executables.
- Common linux commands you need to use in single-user modes are located under this directory.
- Commands used by all the users of the system are located here e.g. `ps`, `ls`, `ping`, `grep`, `cp`



3. /boot :

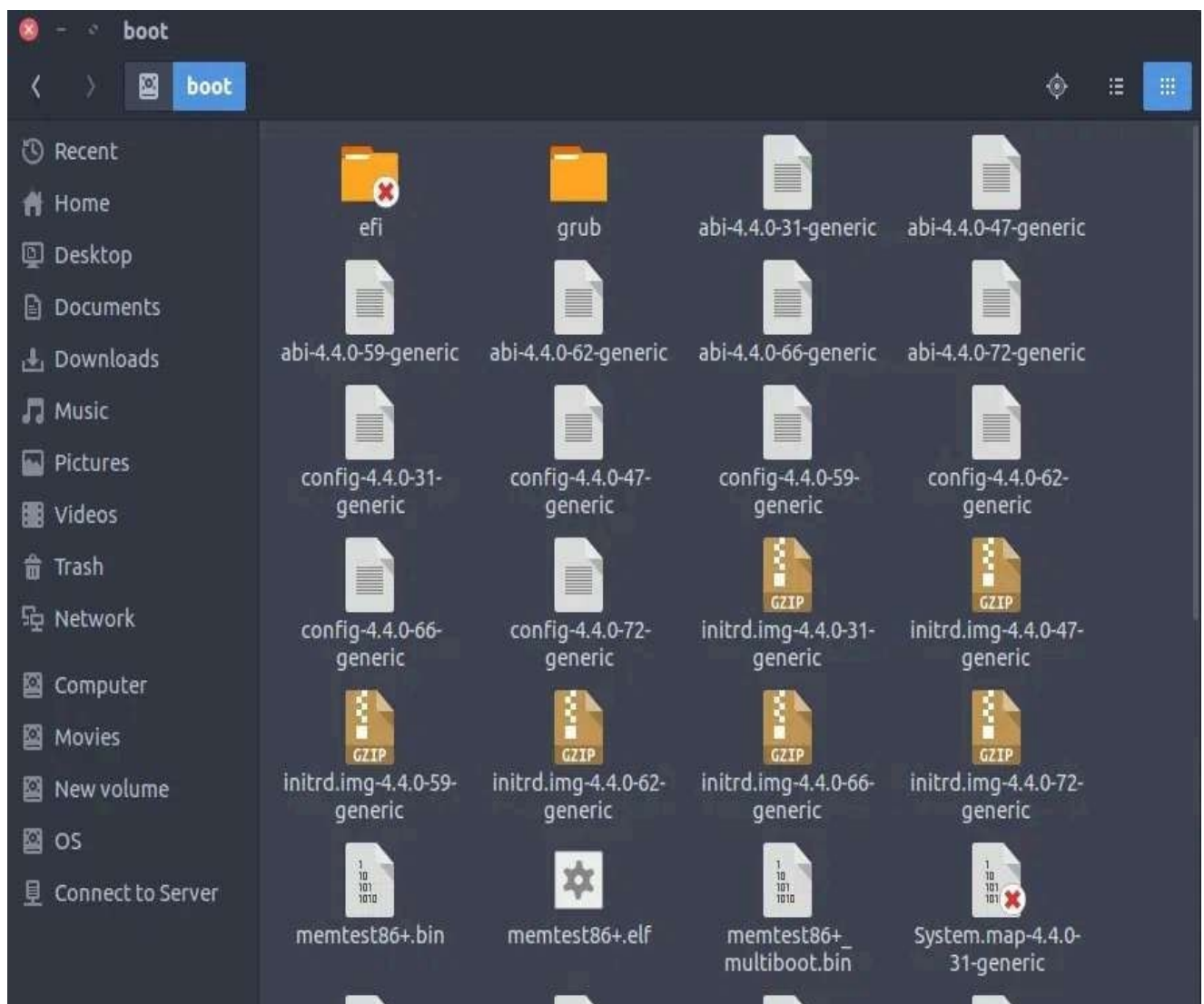
This directory stores [all files required for booting the system](#).

It includes [the GRUB bootloader configuration](#) and [essential kernel files](#) that are loaded during startup.

- Kernel initrd, vmlinuz, grub files are located under /boot
- Example: initrd.img-2.6.32-24-generic, vmlinuz-2.6.32-24-generic

Note: GRUB is the [default bootloader for most Linux distributions](#).

It [presents a menu at startup where users can choose which OS/kernel to boot](#) (useful in dual-boot systems), and it passes control to the selected OS.



4. /dev :

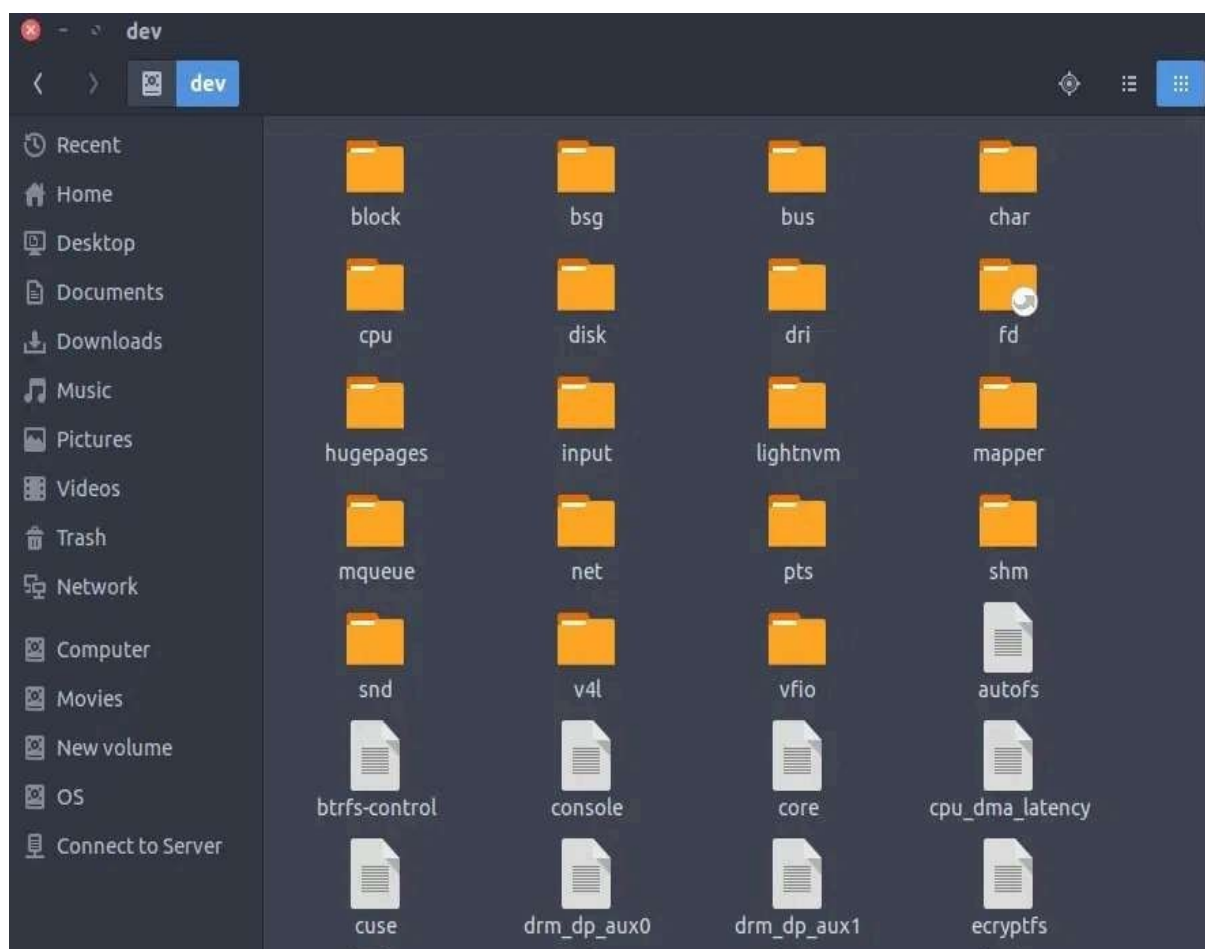
Device files in Linux are stored in the /dev directory.

These are [special files that act as interfaces between hardware and software](#).

Device files are of [two types](#): **block devices** (e.g., hard drives) and **character devices** (e.g., microphones and speakers).

Examples include **/dev/sda1** for disk partitions.

- These include terminal devices, usb, or any device attached to the system.
- Example: /dev/tty1, /dev/usbmon0



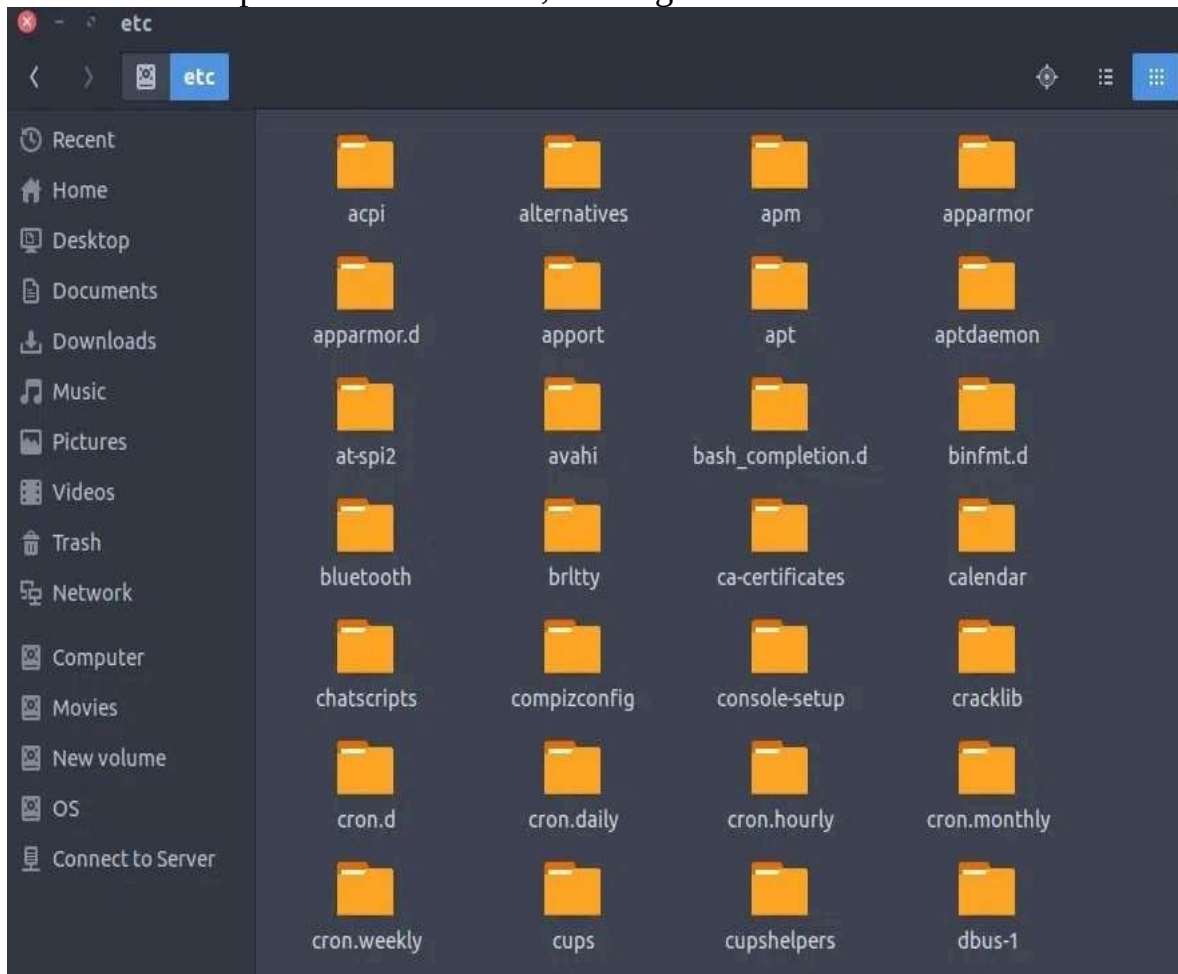
5. /etc :

Short for "[Editable Text Configuration](#)," /etc contains [configuration files for system applications, users, services, and tools](#) or it contains the Host-specific system-wide configuration files.

For example, user [details like UID and local addresses are defined here](#).

- This [also contains startup and shutdown shell](#) scripts used to start/stop individual programs.

- Example: /etc/resolv.conf, /etc/logrotate.conf.



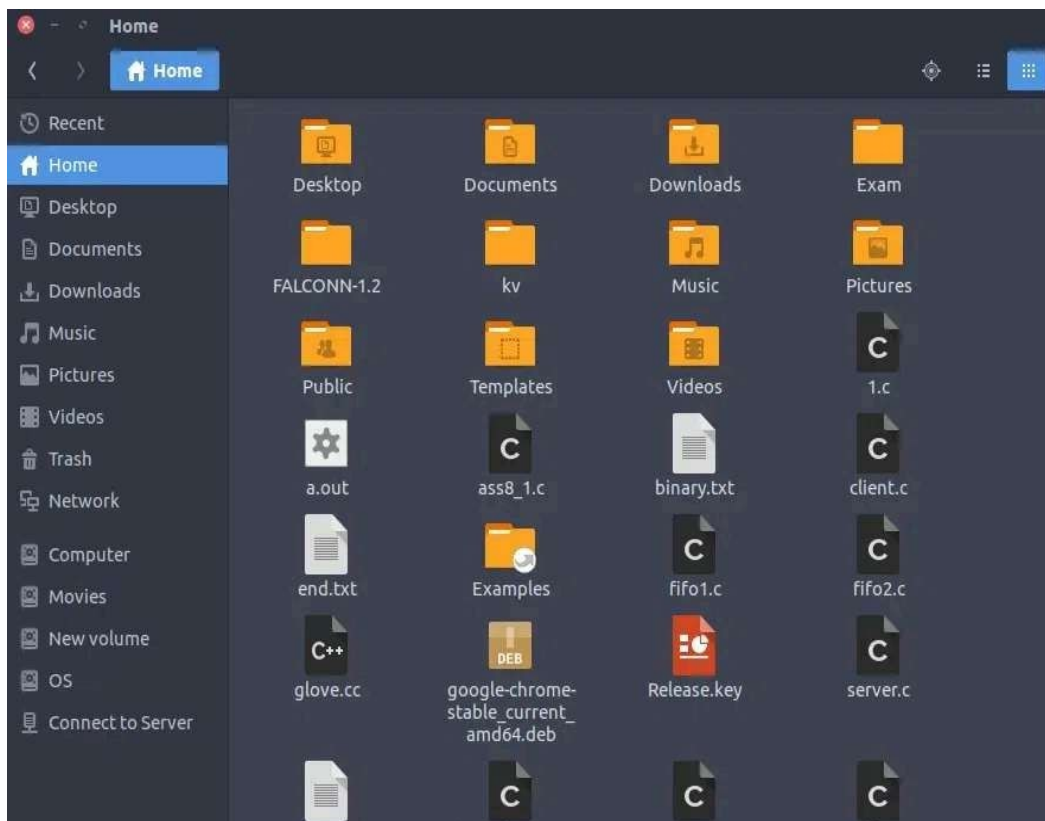
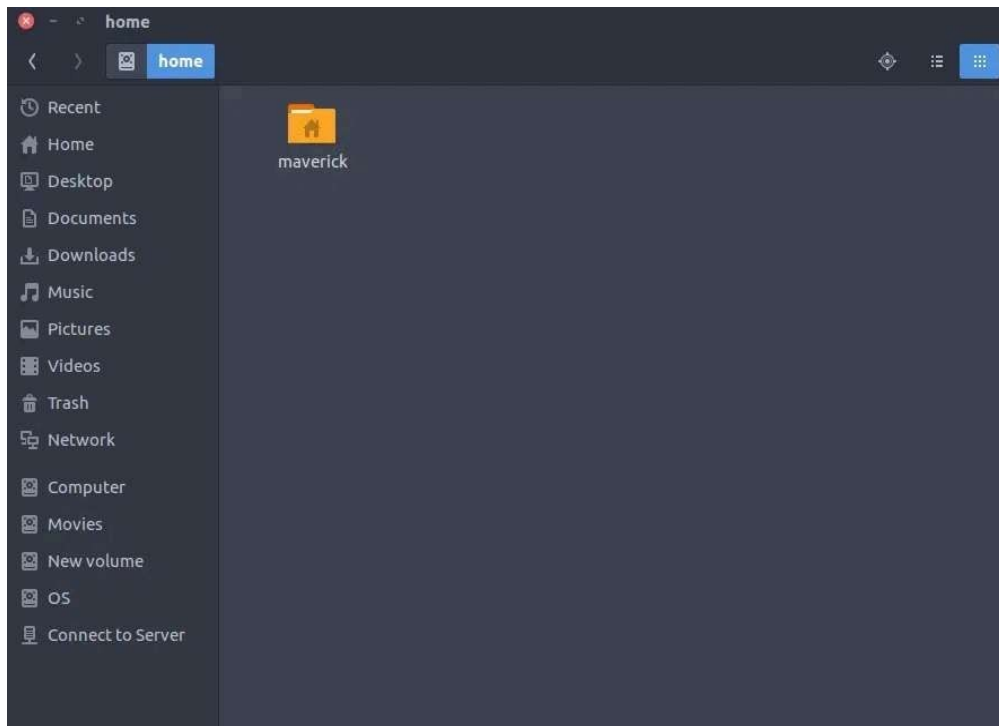
6. /home :

Every non-root user has a personal directory inside /home.

For example, if your username is johndoe, your personal directory would be /home/johndoe.

Each user **can create, delete, or modify files** only in their own home directory and **cannot access others' home directories**.

- Home directories for all users **to store their personal files, containing saved files, personal settings, etc..**
- example: /home/bca, /home/kv

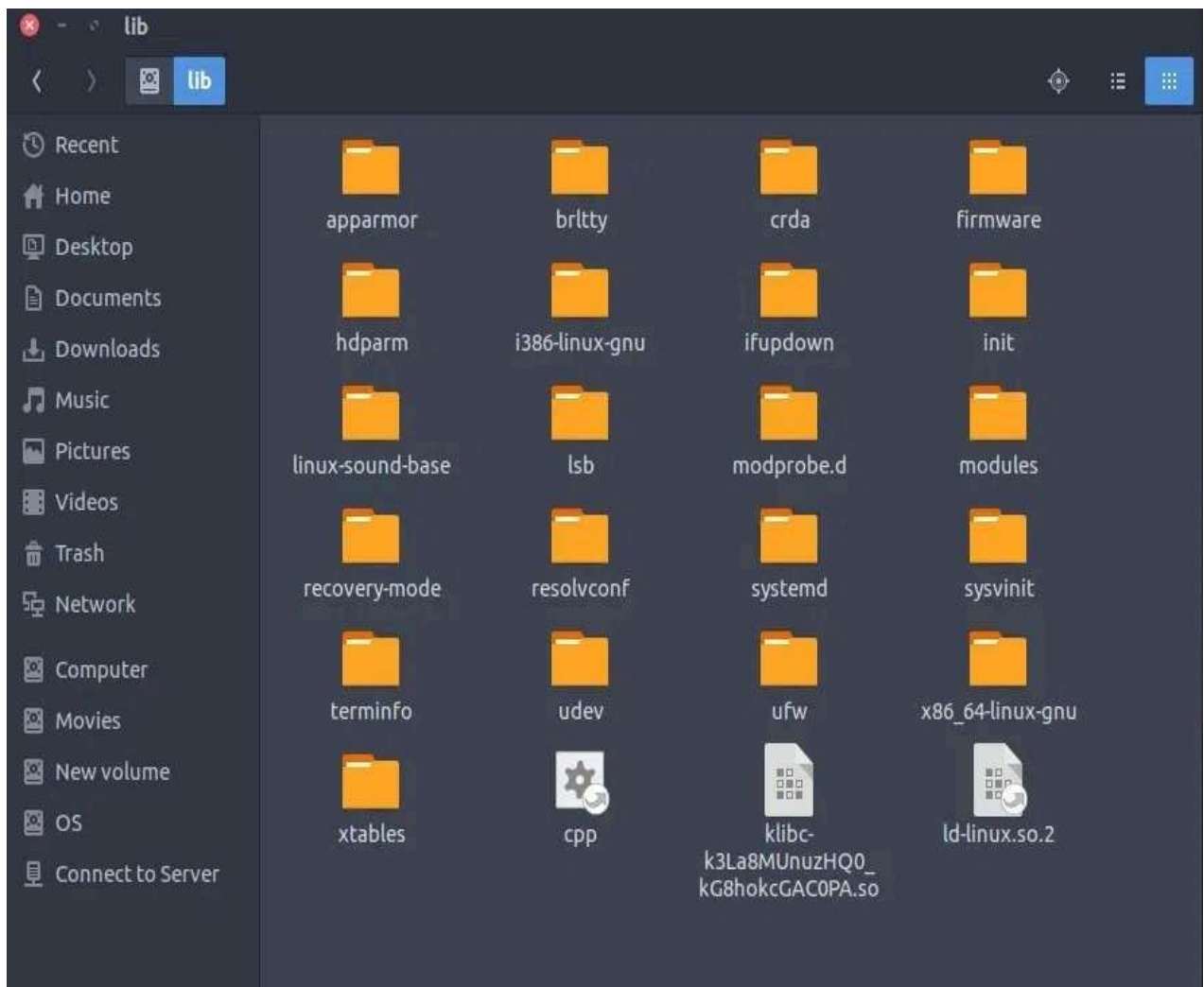


7. /lib:

Applications require shared libraries to run, which are stored in /lib. These include dynamic libraries needed during runtime.

For example, Apache server libraries are available here.

- Library filenames are either ld* or lib*.so.*
- Example: ld-2.11.1.so, libncurses.so.5.7

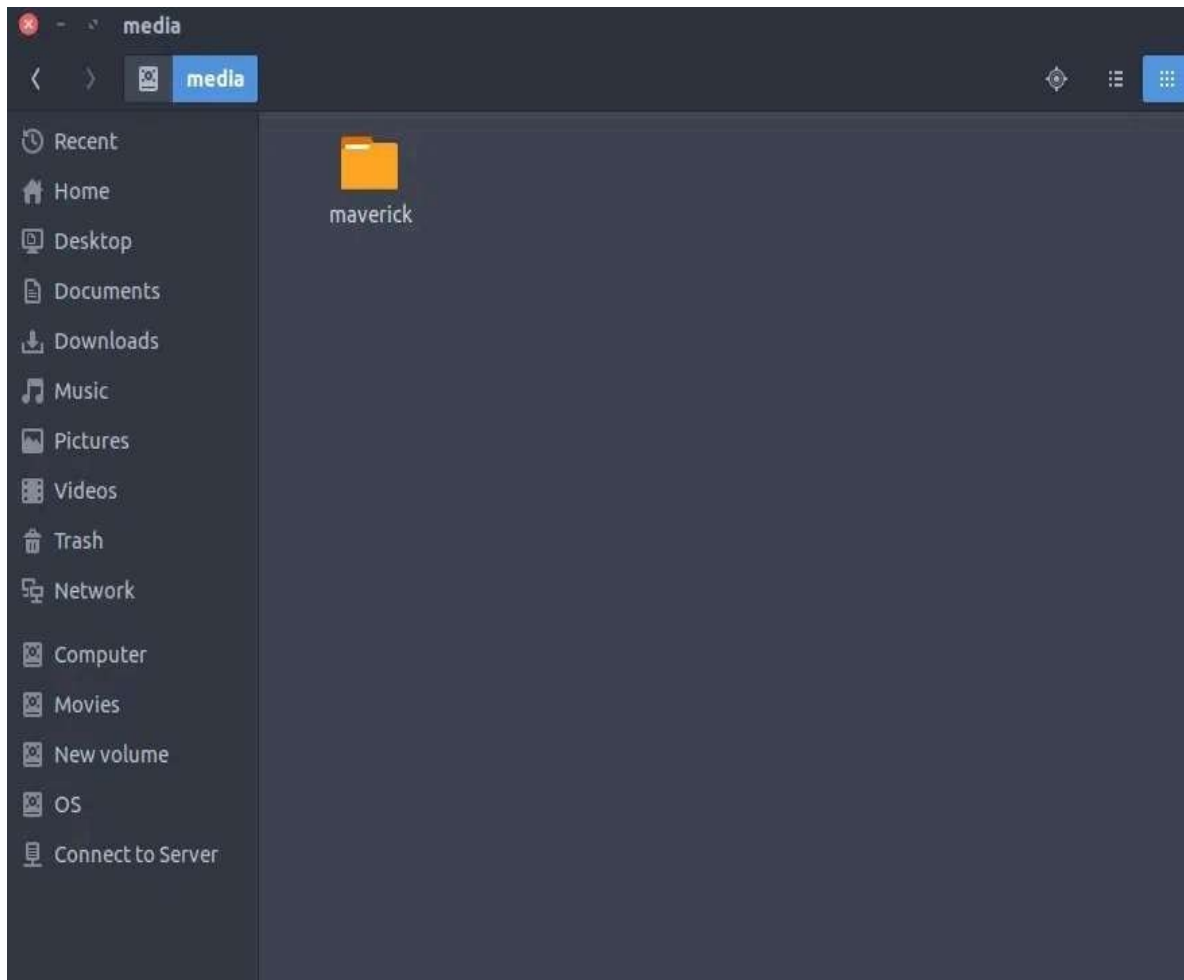


8. /media:

Devices like USBs, CDs, and pen drives are mounted under /media.

For example, when a CD-ROM is inserted (appeared in FHS-2.3), its details will appear here.

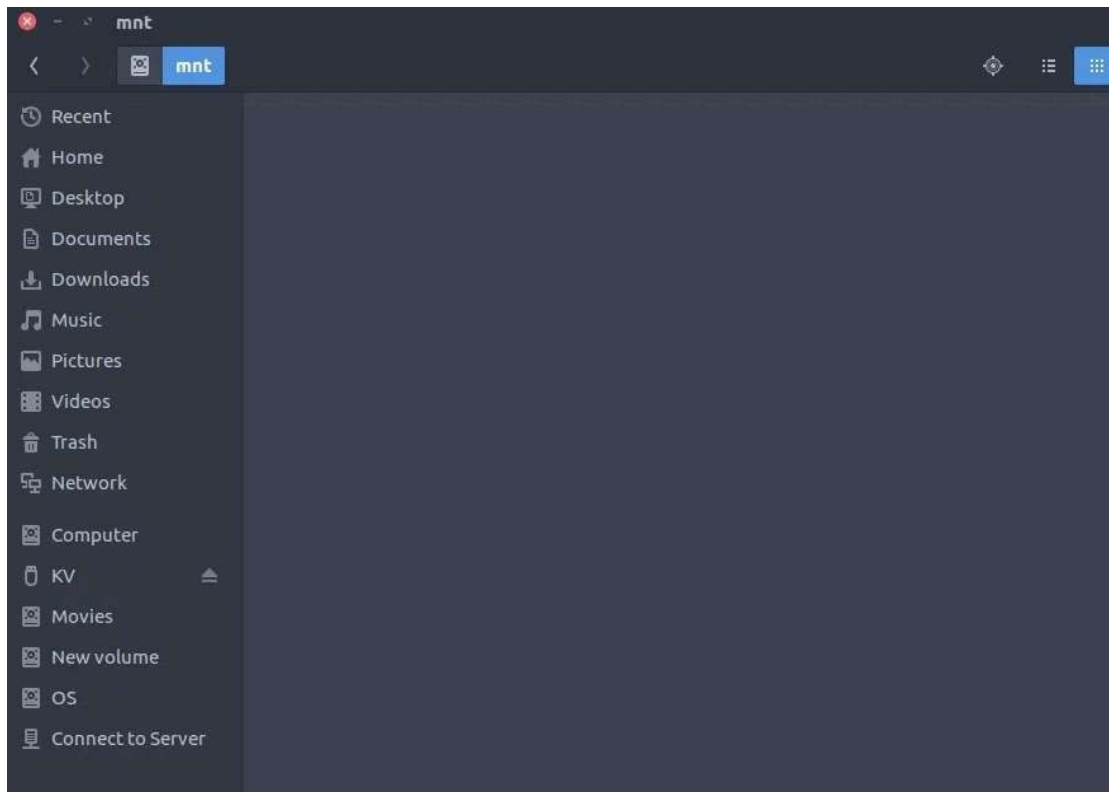
- Temporary mount directory for removable devices.
- Examples, /media/cdrom for CD-ROM; /media/floppy for floppy drives; /media/cdrecorder for CD writer



9. /mnt :

When external drives are connected, they are temporarily mounted in /mnt. This is where their contents become accessible to the system.

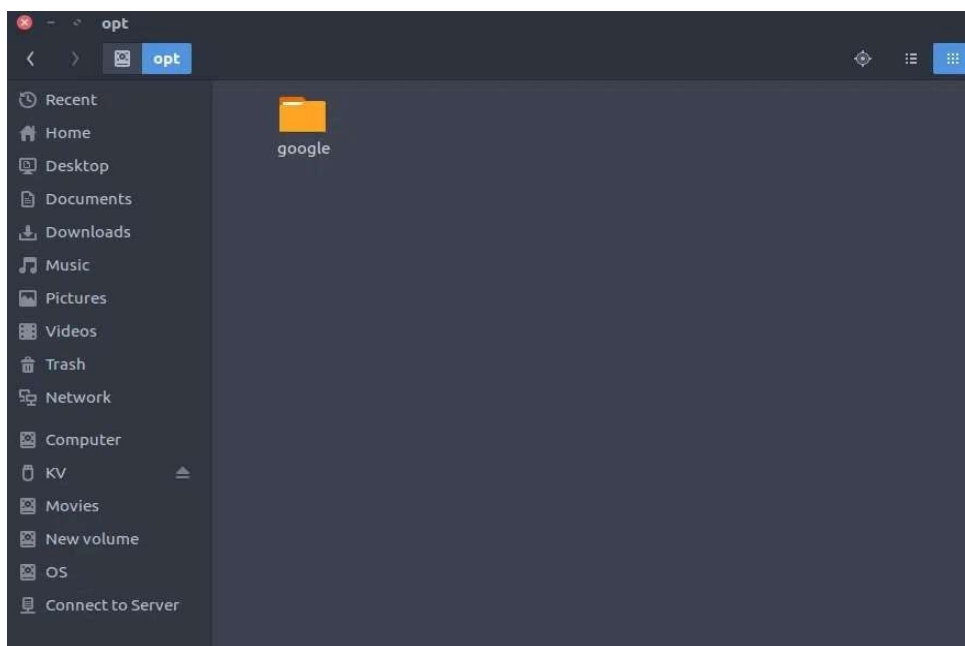
- Temporary mount directory where sysadmins can mount filesystems.



10. /opt :

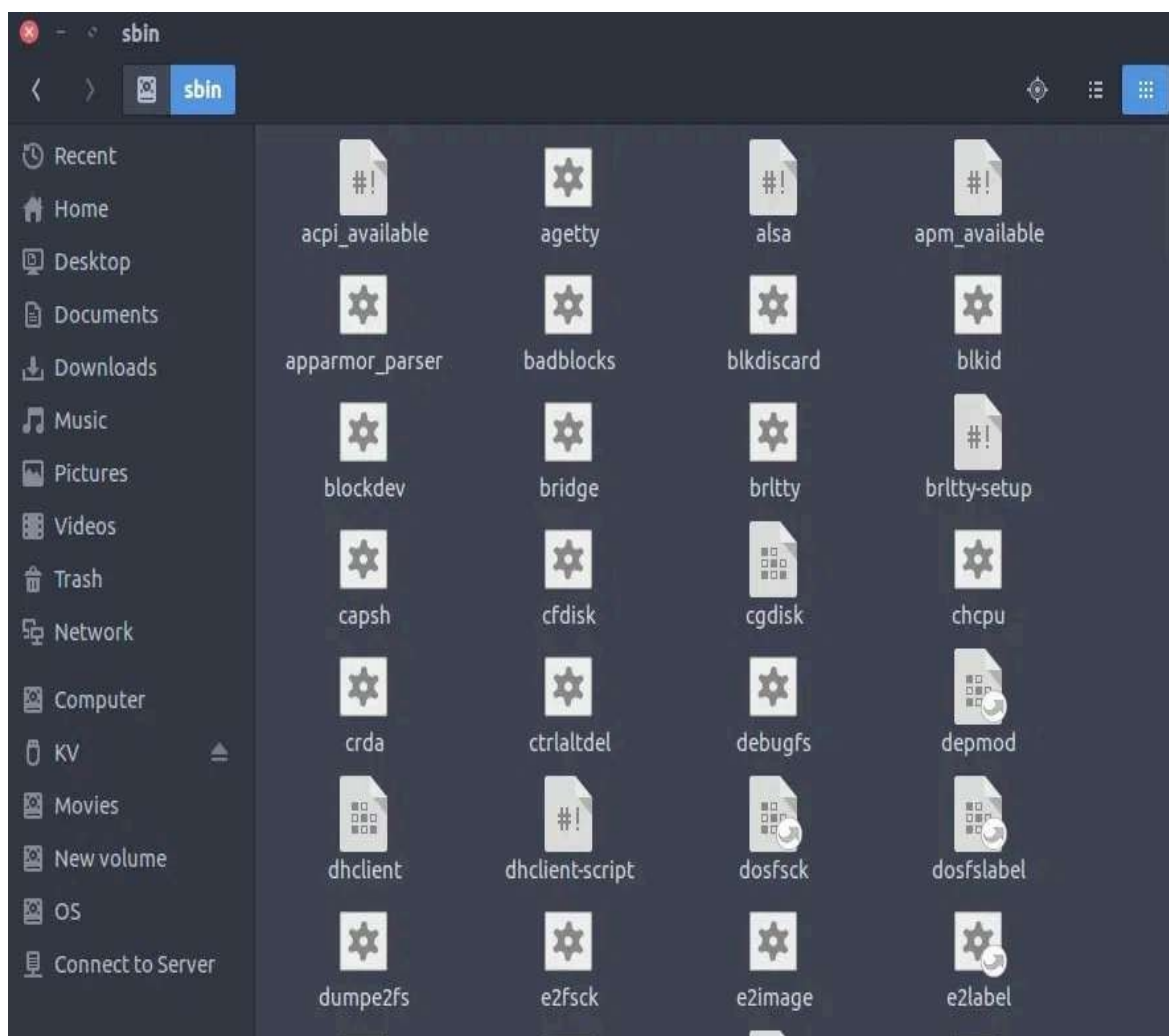
Third-party software and packages not part of the default system installation are stored in /opt. It includes their configuration and data files.

- Contains add-on applications from individual vendors.
- Add-on applications should be installed under either /opt/ or /opt/ sub-directory.



11. /sbin :

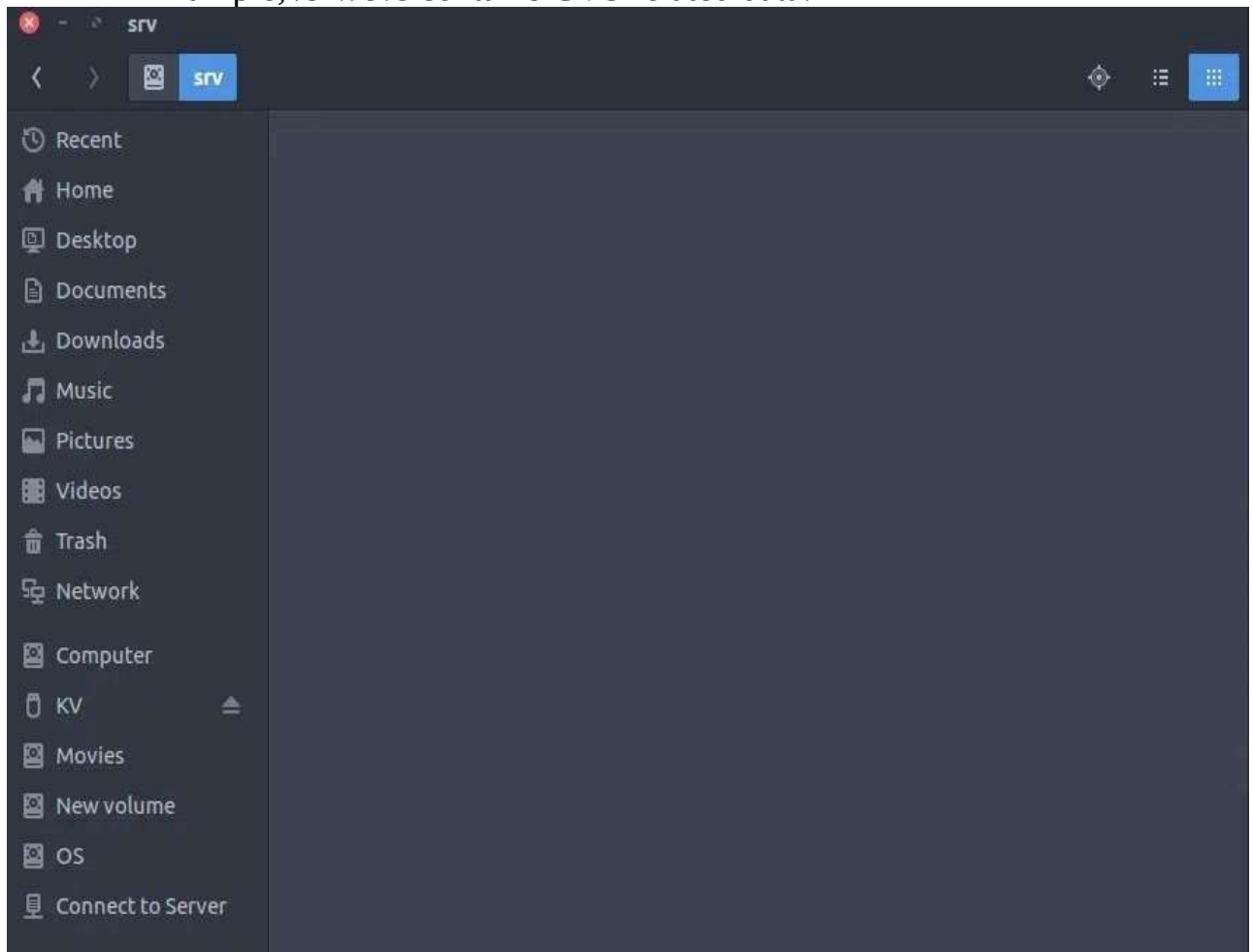
- Essential system binaries.
- This directory holds administrative binaries like iptables, firewall management tools, fsck, init, route etc.
- These binaries are primarily for system administrators and typically require root privileges to execute.
 - Just like /bin, /sbin also contains binary executables.
 - The linux commands located under this directory are used typically by system administrators, for system maintenance purposes.
 - Example: **iptables, reboot, fdisk, ifconfig, swapon**



12. /srv :

Site-specific data served by system, such as data and scripts for web servers, data offered by FTP servers, and repositories for version control systems.

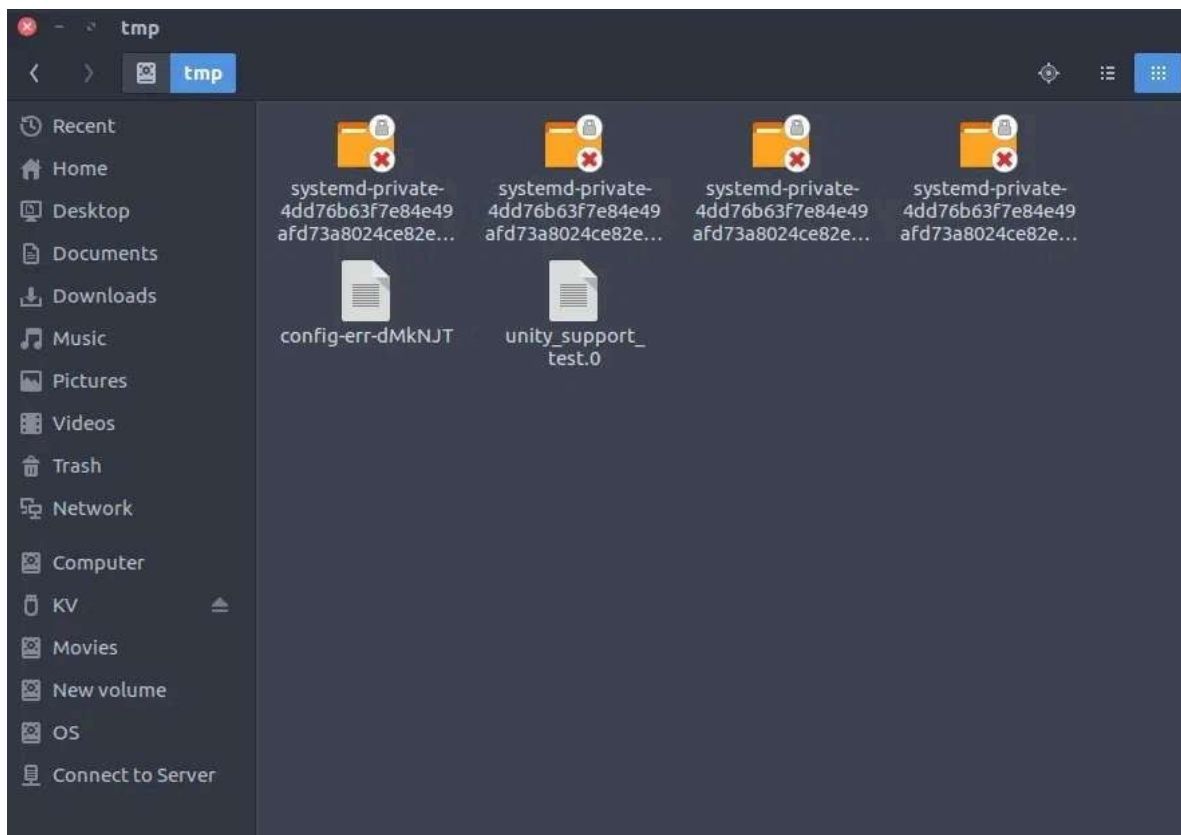
- `srv` stands for **service**.
- Contains server specific services related data.
- Example, `/srv/cvs` contains CVS related data.



13. `/tmp` :

Programs create temporary files during execution, and these are stored in `/tmp`. These files are deleted automatically after the program finishes or when the system is restarted.

- Directory that contains temporary files created by system and users.
- Files under this directory are deleted when the system is rebooted.

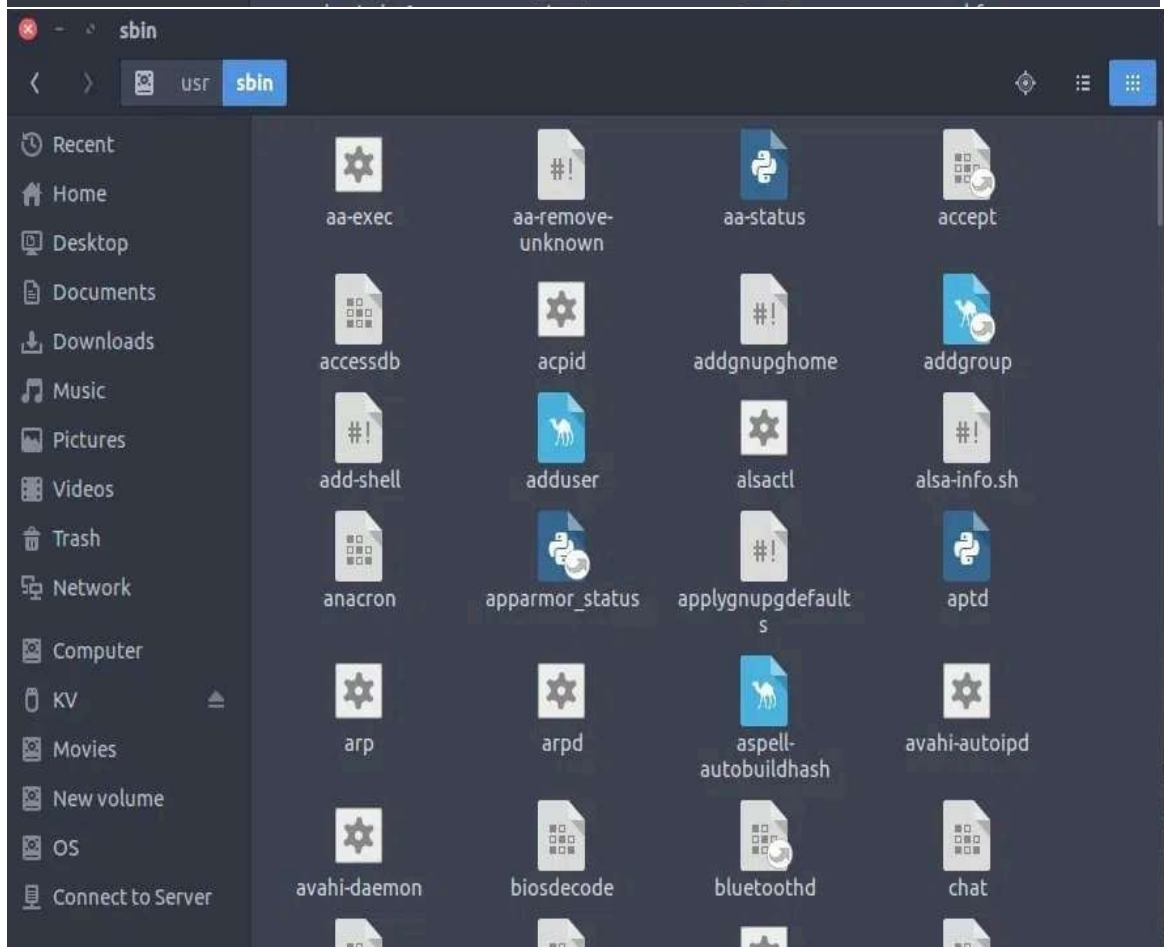
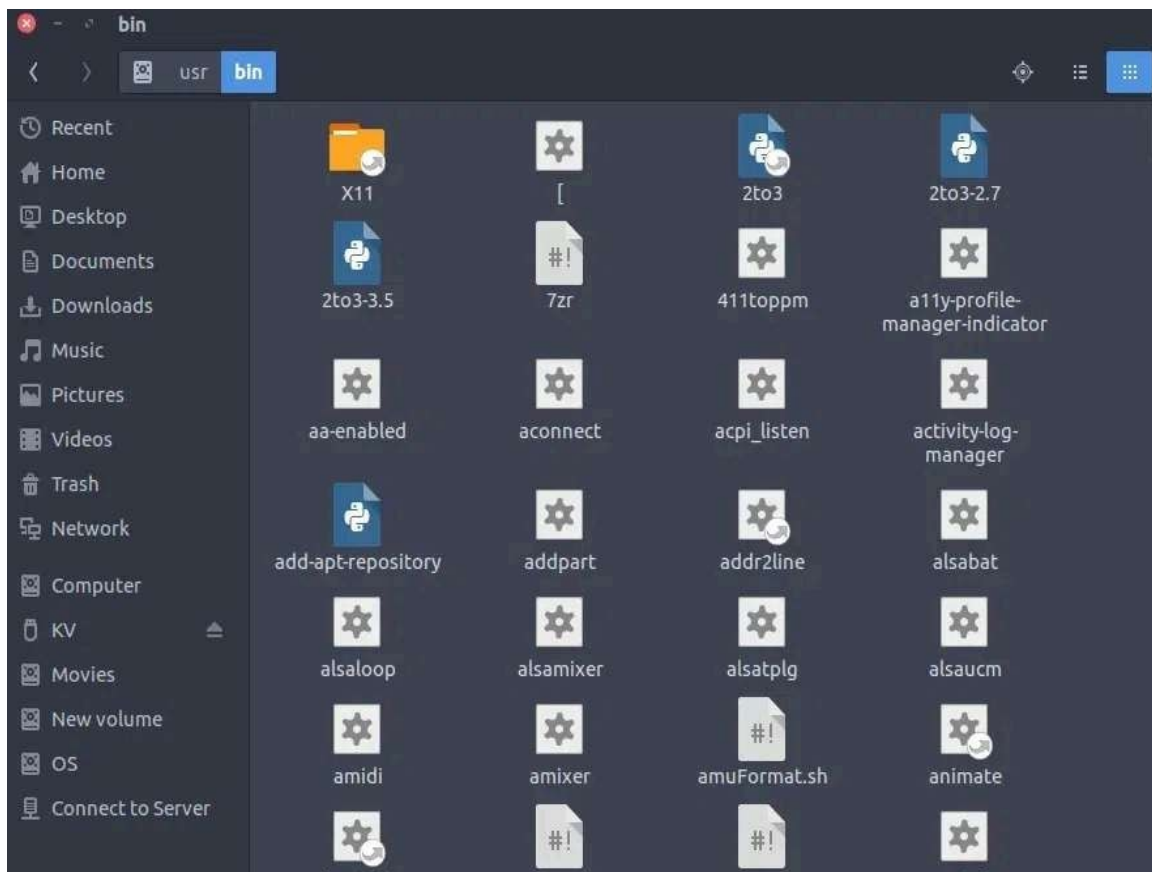


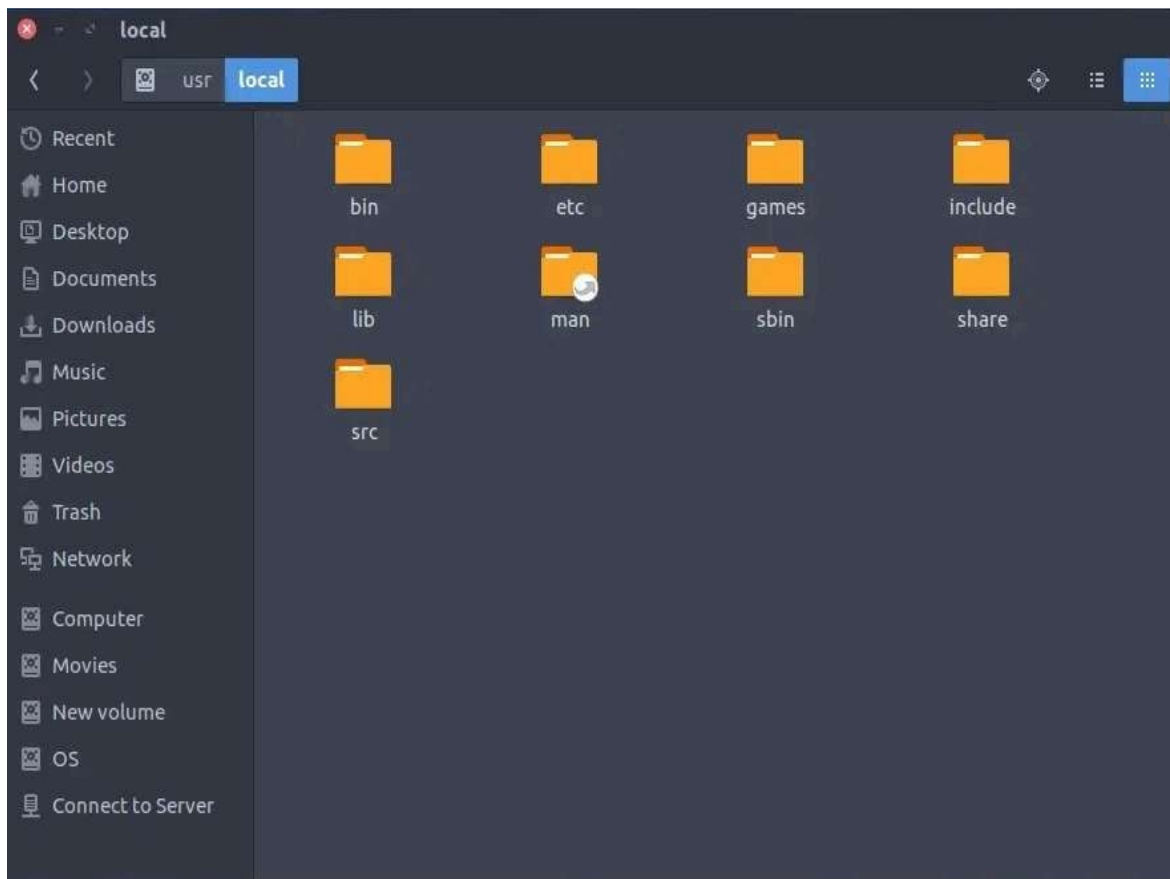
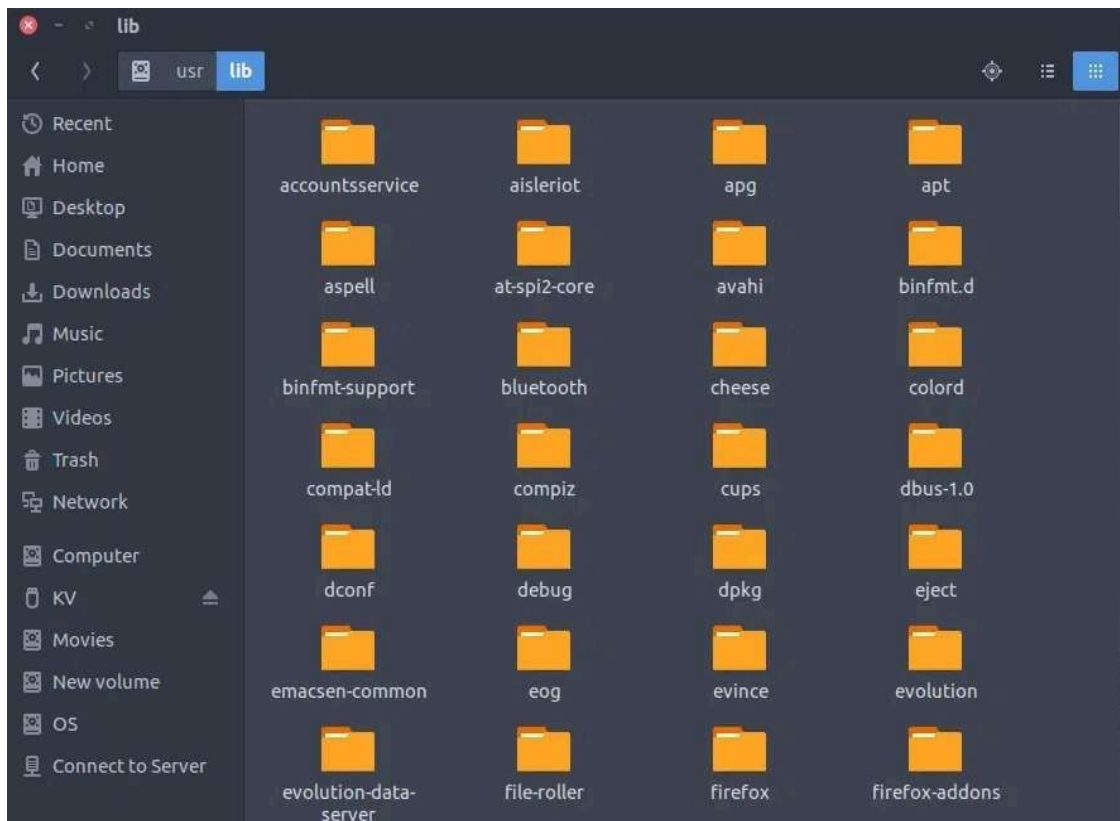
14. /usr :

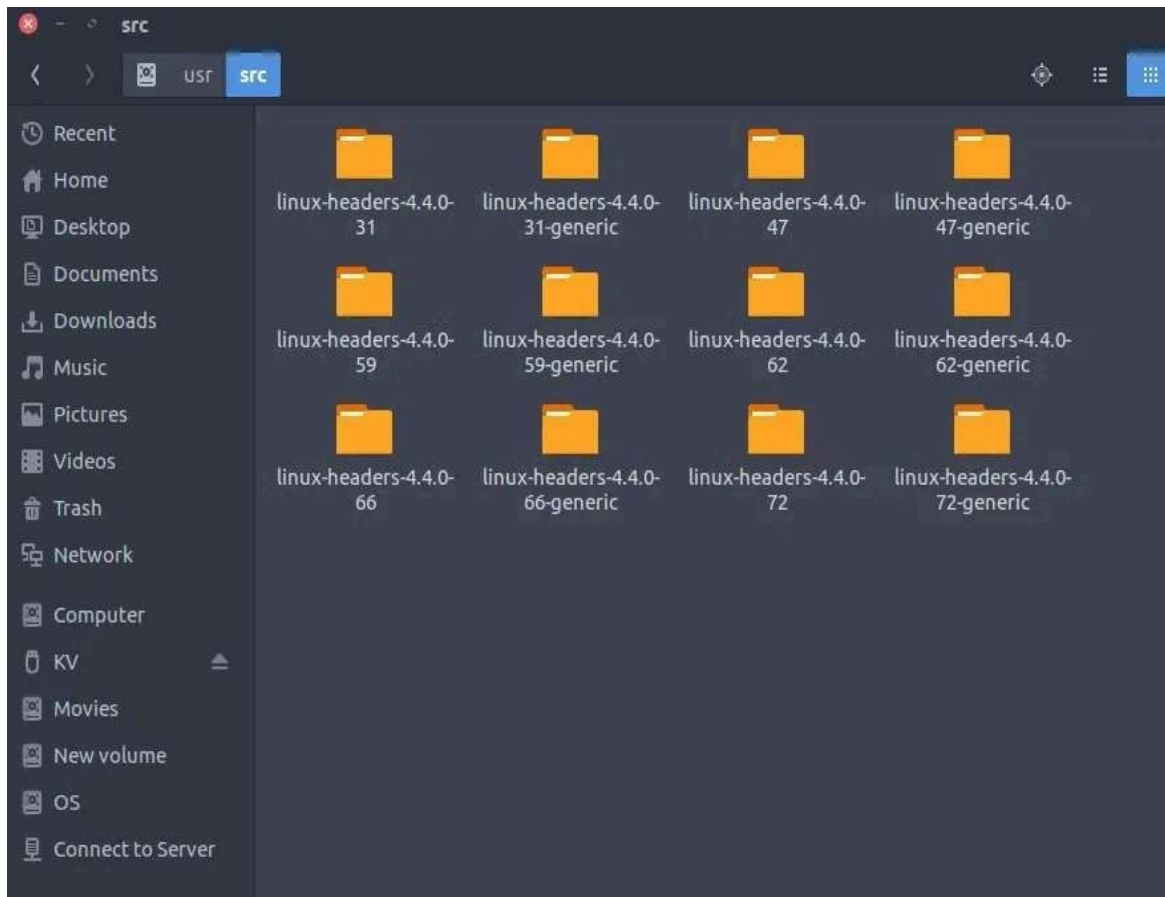
Secondary hierarchy for read-only user data;

Contains the majority of (multi-)user utilities and applications.

- Contains binaries, libraries, documentation, and source-code for second level programs.
- /usr/bin contains binary files for user programs. If you can't find a user binary under /bin, look under /usr/bin. For example: at, awk, cc, less, scp
- /usr/sbin contains binary files for system administrators. If you can't find a system binary under /sbin, look under /usr/sbin. For example: atd, cron, sshd, useradd, userdel
- /usr/lib contains libraries for /usr/bin and /usr/sbin
- /usr/local contains user's programs that you install from source. For example, when you install apache from source, it goes under /usr/local/apache2
- /usr/src holds the Linux kernel sources, header-files and documentation.





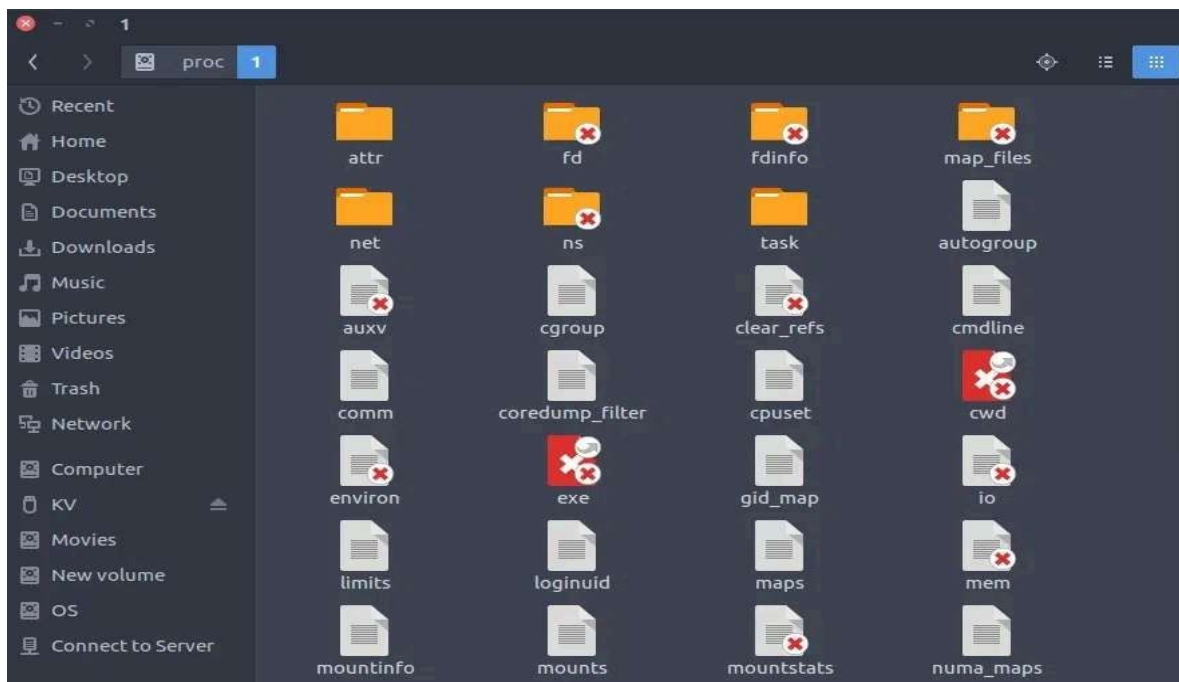
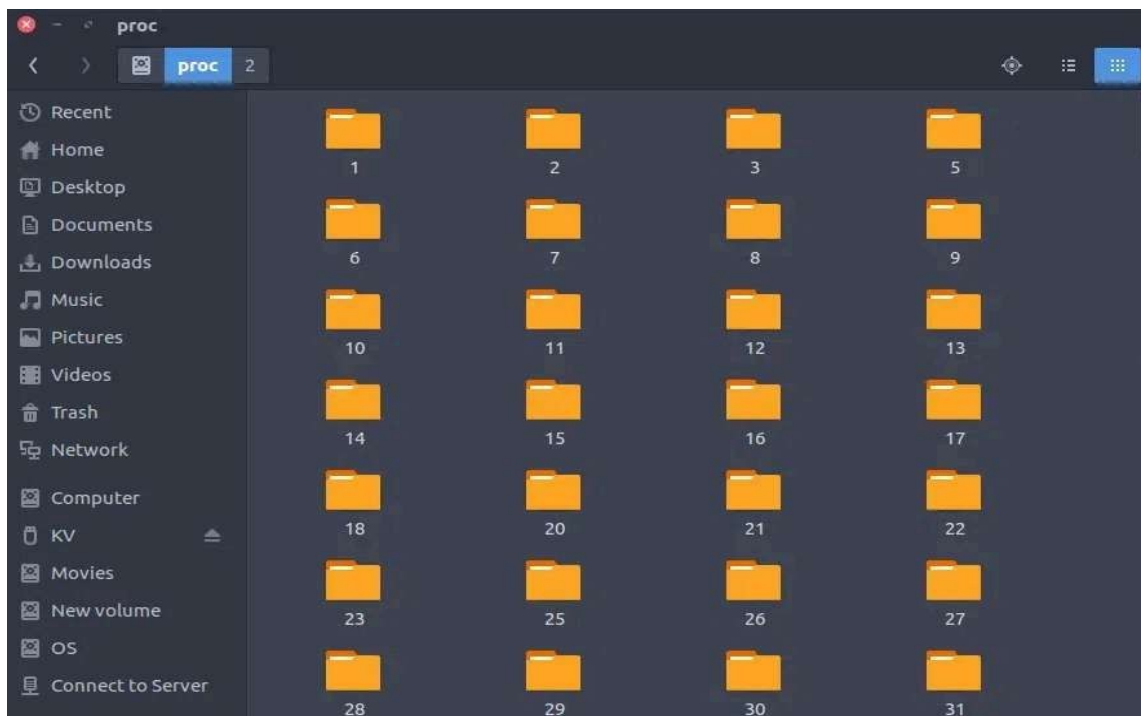


15. /proc:

The /proc directory provides detailed information about system processes. Each process is assigned a unique ID and represented as a directory inside /proc.

For example, /proc/meminfo gives real-time data about memory usage including total, free, buffer, and cache statistics.

- Contains information about system process.
- This is a pseudo filesystem that contains information about running processes. For example: /proc/{pid} directory contains information about the process with that particular pid.
- This is a virtual filesystem with text information about system resources. For example: /proc/uptime



The Linux File System (FHS) to store everything like [programs and settings](#) to [your personal files](#). It starts at the [root folder \(/\)](#) and [organizes files into special folders with specific jobs](#). Key folders include /home for your personal files, /etc for system settings, /bin for essential commands, and /tmp for temporary files. This organization keeps Linux systems tidy, secure, and consistent across different computers.